



AI ENGINEERING, DEMYSTIFIED

10

CONFUSING LLM CONCEPTS

explained simply

The terms every AI engineer hears — but few can explain clearly.

On vs Off-policy

Pre vs Post-training

CPU/GPU/TPU

Quantization



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Swipe →

On-Policy vs Off-Policy Learning

• On-Policy

Learns from its own outputs.

- Model solves 100 problems
- A reward model scores them
- High-scoring answers reinforced

EXAMPLES

RLHF

GRPO

PPO

Like a student reviewing their own exam papers.

• Off-Policy

Learns from someone else's answers.

- GPT-5 generates solutions
- A smaller model learns from them

EXAMPLES

Distillation

Teacher-Student

Like a student learning from a teacher's notes.



Pretraining vs Mid vs Post-training

● Pretraining

Learns language & world knowledge.

- Web pages
- Books
- Code repos

OUTPUT

Base model

● Mid-training

Specializes the model.

- Medicine
- Law
- Coding
- Long-context

● Post-training

Turns it into an assistant.

METHODS

SFT

RLHF

RLVR

School → College → Job training



Zero vs One vs Few-shot Prompting

● Zero-shot

No examples given.

Translate to
French:
"Good morning"

● One-shot

One example first.

Hello → Bonjour
Good morning →
?

● Few-shot

Several examples first.

Hello → Bonjour
Good night →
Bonne nuit
Thank you →
Merci
Good morning →
?

More examples generally improve consistency.





CPU vs GPU vs TPU

● CPU

Few powerful cores.

BEST FOR

- API calls
- Tool execution
- Scheduling
- Orchestration

● GPU

Thousands of parallel cores.

BEST FOR

- Matrix multiply
- Training
- Inference

● TPU

Google's custom AI hardware.

BEST FOR

- Distributed training
- JAX workloads

CPU → Project manager · GPU → Workers · TPU → Specialized factory



Words vs Tokens

- **Words**

What humans understand.

- **Tokens**

What LLMs understand.

EXAMPLE

“unbelievable”



un

believ

able

TOKEN COUNT DETERMINES

- Cost
- Context size
- Latency

RULE OF THUMB

100 words ≈ 130 tokens



Logits vs Probabilities

The model doesn't predict words directly. It generates raw scores, then softmax normalizes them.

• ① Logits

Raw confidence scores.

Dog	8.2
Cat	5.1
Car	-1.4

• ② Probabilities

Normalized by softmax.

Dog	92%
Cat	7%
Car	1%

Logits = raw confidence · Probabilities = normalized confidence.





Context Window vs Memory

● Context Window

Temporary working memory.

SIZES

32K

128K

1M tokens

Ends with the conversation – gone forever.

● Memory

Persistent across sessions.

TYPES

- Semantic
- Episodic
- Procedural

Context Window = RAM · Memory = Hard disk



LLM vs AI Agent

• LLM

Input → Output.

- No tools
- No memory
- No actions

• AI Agent

Input → Plan → Tools → Observe
→ Repeat.

- Search the web
- Run code
- Access databases
- Use memory & workflows

Agent = LLM + Tools + Memory + Loop



Inference vs Decoding

• Inference

The whole generation process.

STAGES

- 1 · Prefill
- 2 · Decode

• Decoding

Choosing the next token.

STRATEGIES

- Greedy
- Beam search
- Top-K
- Top-P

Temperature controls the randomness.

Decoding is just one part of inference.



Pruning vs Quantization

- **Pruning**

Remove unnecessary parameters.

70B → 60B

Smaller network.

- **Quantization**

Keep all parameters, lower precision.

FP16 → INT8 → INT4

Same network, smaller numbers.

Pruning → remove books · Quantization → print in a smaller font



Using LLMs is easy. Understanding them isn't.

You use these terms daily:

RLHF

Distillation

Quantization

Context Windows

Inference

Understanding them changes how you:

✓ Optimize latency

✓ Reduce costs

✓ Debug hallucinations

✓ Build production AI

Understanding how LLMs work is where AI engineering actually begins.



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